Raspberry PI Pico: The Wonder Board for IoT Applications

This hands-on tutorial with a Raspberry Pi Pico microcontroller board will get you started on your IoT journey.

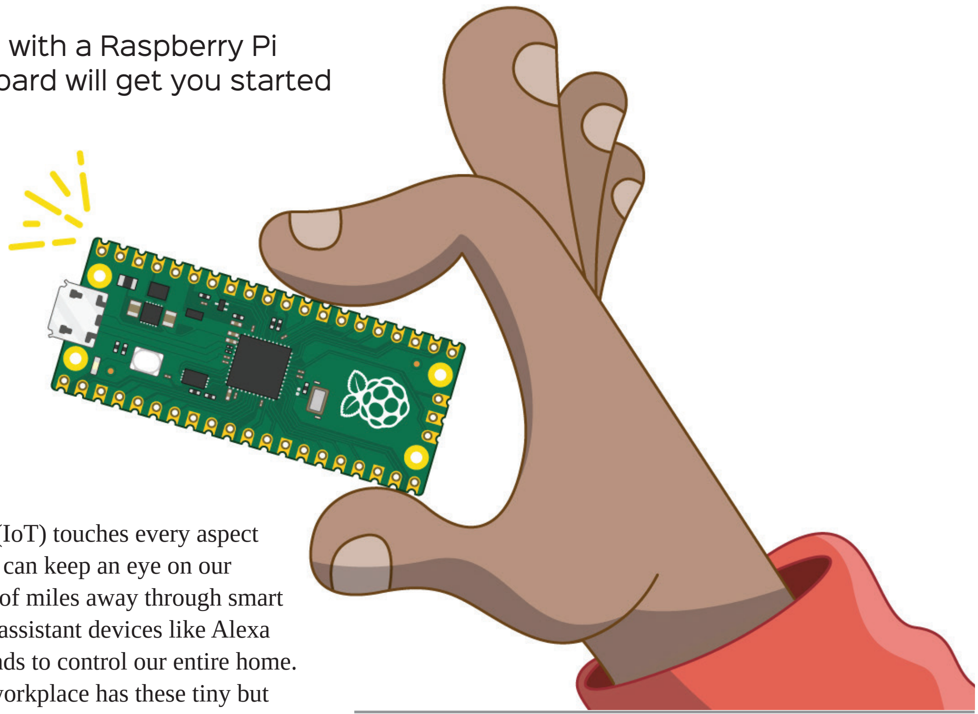

Image Source: *https://projects.raspberrypi.org*

The Internet of Things (IoT) touches every aspect of our lives today. We can keep an eye on our home from hundreds of miles away through smart security cameras, while home assistant devices like Alexa obediently follow our commands to control our entire home. Every corner of our home or workplace has these tiny but powerful intelligent devices that create a whole automated system to provide a comfortable life.

Recent advances in open hardware and software standards-based systems have helped replace pricey and complicated embedded systems. The Raspberry Pi (RPI) Foundation has designed many general-purpose single board, low power and cost-effective hardware systems in the last few years to help create powerful IoT systems with a fraction of the effort and cost involved earlier. Millions of intelligent devices around the world are already using RPI hardware for modern IoT usage. So let's jump onto this rocket ship of IoT too.

The tools and examples used in this brief tutorial are tested on Ubuntu 18.04 LTS on an AMD64 laptop and the latest Raspberry PI OS on RPI4, respectively. But everything should work on macOS and Windows too as the official tools are provided for those as well.

First encounter with PI Pico

Raspberry PI Pico 1 / 2 are generation 1 and 2 families of microcontroller boards with system-on-chip (SoC) based on ARM dual core RP 20240/2350 processors. Both families have boards with wireless LAN and Bluetooth connectivity too. You should be able to buy a Pico 1 board from a Raspberry Pi reseller in your country at a price of roughly ₹ 300 to ₹ 400 through the Raspberry Pi official website.

Remember that when you buy a board through reseller websites, if you see a letter H added after Pico 1 / 2 it means the header is soldered on the board. The board with header could be inserted at various places to control and interact with more external hardware components. And when you see the letter W appended after your Pico 1 / 2 series it indicates boards with wireless capabilities. The Pico 2 family is just an upgrade over Pico 1, with roughly double the RAM and faster cores, etc.

We'll use the Pico 1 series board in this article to get started with the necessary fundamentals to quickly embark on our IoT journey. We don't need any external electronic component, except a micro USB to USB cable to experiment with the Pico code snippets given in this section. Your Pico board should look like what's shown in Figure 1 with important components like a USB port to program/power, onboard LED, BOOTSEL button, and a big black microcontroller chip in the middle clearly visible on it. Don't worry if you can't understand a few of the terms used in this section. Just follow along and you'll be able to correlate with most of the things.